

Mobile App Development 1

Computing and Mathematics

May, 2026

Mobile App Development 1 (A13663)

Short Title:	Mobile App Development 1	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Software and Web Development	
Author	EDELEASTAR	
Credits	5	Level: Advanced

Description of Module / Aims

Design, build and deploy a multi-screen mobile application incorporating an intuitive and efficient navigation mechanism. Structure the implementation using accepted best-practice with respect to patterns, frameworks and tools. Incorporate localised persistence models + simple access to remote services. Introduce context services such as location/camera and/or other sensor access.

Programmes

COMP-0630	BSc (Hons) in Applied Computing (International) (WD_KACMI_B)	4 / 7 / M
COMP-0630	BSc (Hons) in Applied Computing (WD_KACCM_B)	4 / 7 / M
COMP-0630	BSc (Hons) in Applied Computing (WD_KCOMP_B)	4 / 7 / M
COMP-0630	BSc (Hons) in Computer Forensics and Security (WD_KCOFO_B)	4 / 7 / M
COMP-0630	BSc (Hons) in Computer Science (WD_KCMSC_B)	4 / 7 / M
COMP-0630	BSc (Hons) in Creative Computing (WD_KCRCO_B)	4 / 8 / M
COMP-0630	BSc (Hons) in Multimedia Applications Development (WD_KMULM_B)	4 / 1 / M
COMP-0630	BSc (Hons) in Software Engineering (WD_KDEVP_BI)	3 / 6 / M
COMP-0630	BSc (Hons) in Software Systems Development (WD_KDEVP_B)	3 / 6 / M
COMP-0630	BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI)	3 / 6 / M
COMP-0630	BSc in Applied Computing (WD_KCOMP_D)	3 / 6 / M
COMP-0468	BSc in Information Technology (WD_KINFT_D)	2 / 4 / M
COMP-0630	BSc in Software Systems Development (WD_KCOMC_D)	3 / 6 / M

Indicative Content

- Application Structure: Components; Resources; Security; General Assets
- User Experience: UX Principles, Navigation, Imagery, Fonts
- Simple User Interaction Patterns
- Essential Application Structure Patterns: Appropriate Variations on Model/View/Controller (MVVM, MVP etc\ldots)
- Resource access and management; Clean separation of concerns
- Application Life-cycle: Startup/shutdown; Foreground/background
- UI State Preservation and Restoration; Concurrency

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Decompose an application into its constituent parts, including but not limited to: core application components, user experience resources, packaging.
2. Design a coherent User Experience - using appropriate tools, practices and guidelines - for a moderately sized application.
3. Produce medium sized application, based on a limited set of design patterns.
4. Manage the application lifecycle.
5. Structure persistent storage on a device and reliably save and restore application state.

Learning and Teaching Methods

- Lectures will introduce the general context of the curriculum, and explore specific topics in depth.
- Supervised, guided and scripted practicals will lead the student through the construction of an application designed to illustrate key concepts covered in the lectures.
- The focus is on learning by doing in a studio environment. Each practical will propose a set of exercises -- to be solved in a subsequent practical.
- Assignment One will focus ensuring the student can construct a new application equivalent in style and structure to the guided practical.
- Assessment Two will invite the student to analyse, design and implement a new application.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	50%	1,2,5
Assignment	50%	3,4,5

Assessment Criteria

- <40%:** Unable to implement a basic application. Cannot grasp fundamentals of the application lifecycle or operate an appropriate IDE (Integrated Development Environment).
- 40%-49%:** Be able to implement a simple application, with 2-3 separate views/activities.
- 50%-59%:** Understand the basic of the application lifecycle and operate an IDE at a basic level. Ability to model and implement an application of moderate complexity -- including > 3 views + a simple persistence mechanism.
- 60%-69%:** Be able to use an IDE competently and debug applications. Be able to implement a reasonably sophisticated application with multiple view / navigation mechanisms. The application will have local persistent storage and be able to interact with a remote service as a basic level (read only say).
- 70%-100%:** All the above to an excellent level. Be able to build an application that can access on device sensors / subsystems (e.g. location or camera).

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	12
Practical	24	12
Independent Learning	87	111

Essential Material

- "Android Developer Resources." <http://developer.android.com>
- "Apple Developer Resources." <http://developer.apple.com/ios>
- "Cordova Developer Resources." <https://cordova.apache.org>

Supplementary Material

- Camden, R. _Apache Cordova in Action_. New York: Manning, 2015.
- Neuburg, M. _iOS 9 Programming Fundamentals with Swift: Swift, Xcode, and Cocoa Basics_. New York: O'Reilly, 2015.
- Phillips, B. _Android Programming: The Big Nerd Ranch Guide_. New York: Pearson, 2015.

Requested Resources

- COMPUTER LAB: BYOD Lab